

Efficient Implementation of Convolution Encoder and Viterbi Decoder

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Abstract— This paper describes about the Convolution Encoder and Viterbi Decoder algorithm. The main aim is to achieve encoding and decoding rate as per WiMAX standard. Here, Convolution Encoder and Viterbi Decoder of code rate $\frac{1}{2}$, constraint length 7; generator polynomial (171,133) has been implemented on EP4SGX70HF35C2 device of Stratix IV family in Altera DE board. Coding style of VHDL is used. The design has been synthesized using Altera Quartus II v11.0 and has been simulated using ModelSim Altera Starter Edition 6.6d. The comparison results show a large improvement in area.

Keywords— Convolution Encoder, VHDL, Viterbi Decoder, Stratix IV.

I. INTRODUCTION

In telecommunication and coding theory digital data transmission is frequently increasing over noisy channels. Channel coding is a technique used for controlling errors in data transmission. There are many techniques and algorithms available for transmission and reception of digital data over noisy channel but Convolution Encoder and Viterbi Decoder are most efficient [1]. It gives improved coding gain and enhanced performance. Convolution Encoding with Viterbi Decoding is a Forward Error Correction technique that is particularly suited to a channel in which the transmitted signal is corrupted mainly by Additive White Gaussian Noise (AWGN) [2]. Forward Error Correction can be mainly classified in two categories: Block codes and convolution codes. Fixed size blocks of bits are called Block codes while convolution codes work on uninformed length blocks of bits. The Convolution Encoder is frequently used in digital communication systems [3].

II. CONVOLUTION ENCODING

A Convolution code is defined by three integers: n , k and K , where the proportion k/n is termed code rate, n is the number of output bits, k is the number of input bits and K is called constraint length. The constraint length signifies the number of k -bit shifts over which a single information bit can influence the encoder output. Figure 1 shows the architecture of Convolution

Encoder for generator polynomial (171,133) of code rate $\frac{1}{2}[6]$, where u_1 and u_2 are first and second code symbols respectively. Output branch word and input bits to be fed to the register depend on the code rate. Here two bit Output branch word and one bit input are fed to register at each clock cycle.

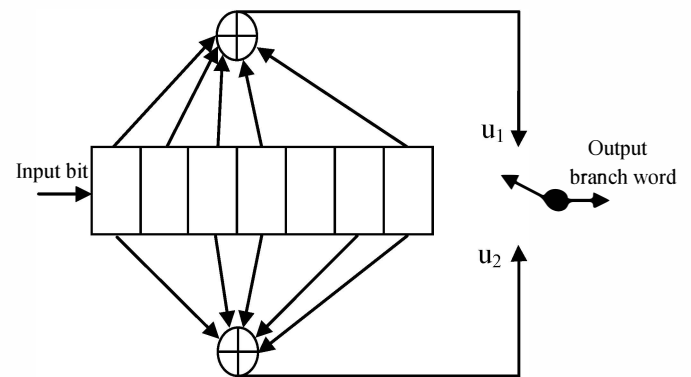


Fig. 1. Architecture of Convolution Encoder for generator polynomial (171,133) of code rate $\frac{1}{2}[6]$.

Let $G_1(x)$ be the generator polynomial for upper connection and $G_2(x)$ be the generator polynomial for lower connection which has been shown in equation (1).

$$\left. \begin{aligned} G_1(x) &= 1 + x_3 + x_4 + x_5 + x_6 \\ G_2(x) &= 1 + x + x_3 + x_4 + x_6 \end{aligned} \right\} \quad (1)$$

In order to calculate u_1 and u_2 XOR operation have to be done as shown in figure 1. The implementation of Convolution Encoder of constraint length 7 and code rate $\frac{1}{2}$ is realized using ModelSim 6.6d and EP4SGX70HF35C2 device of Stratix IV family in Altera DE board.

III. VITERBI DECODING

Viterbi decoder uses the Viterbi algorithm for decoding a data which is encoded using a Convolutional Encoder. Viterbi Decoding algorithm is broadly used in Radio communication, Digital TV, Radio relay, Satellite communication and many

more applications [4]. Figure (2) shows major components of Viterbi Decoder which are branch metric unit (BMU), add compare select unit (ACSU), survivor-path memory unit (SMU), and path metric unit (PMU).

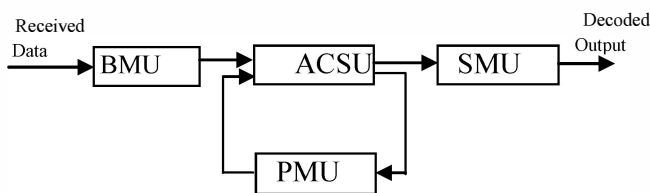


Fig. 2. Functional Diagram of Viterbi Decoder[4].

The branch metrics are calculated from the received channel symbols in BMU which are distances between all the possible symbol codes and data received. Branch matrices are fed into ACS which performs add compare and select operation for all the states. The decision bits produced in ACS are stored and regained from the SMU in order to finally decode the source bits along the final survivor path [4],[5]. The decoded output sequence is determined by tracing back the information from the recorded survivor paths. The core elements of PMU are ACS units. The path metrics of the current iteration are stored in the path metric unit (PMU) and are read out for the use of next iteration [7-9].

In this paper, we propose an area optimized VLSI architecture of Convolution Encoder and Viterbi Decoder. The FPGA implementation of the architecture is done and compared with some existing techniques. The proposed architecture has a large area improvement. The silicon validation of the architecture is done by programming in EP4SGX70HF35C2 device of Stratix IV family in Altera DE board.

The remaining part of this paper is as follows. Section IV describes the flowchart of Viterbi Decoding uses the Viterbi algorithm. Section V contains the implemented results along with simulated outputs, simulated using ModelSim. Conclusion is drawn in section VI.

IV. THE FLOWCHART OF IMPLEMENTED VITERBI DECODER.

The Viterbi Decoding uses the Viterbi algorithm; figure (3) shows all the intermediate steps of the implemented algorithm. It consists of two decision blocks and some internal steps to decode the received data. Branch matrices are calculated from the symbols and the pattern; maximum possibility of BM (Branch matrices) is four. The calculated BM are stored in registers, which have interface with feedback from state ends; all the intermediate steps will be executed for all the possible states.

In intermediate steps, ACS step is used to find the survivor path of the decoding algorithm. Trellis stages consist of the path which is selected by ACS and stored in PMU for all the unit time and states. Decoded output bits are calculated after Trellis stage ends and final survivor path is selected by the algorithm.

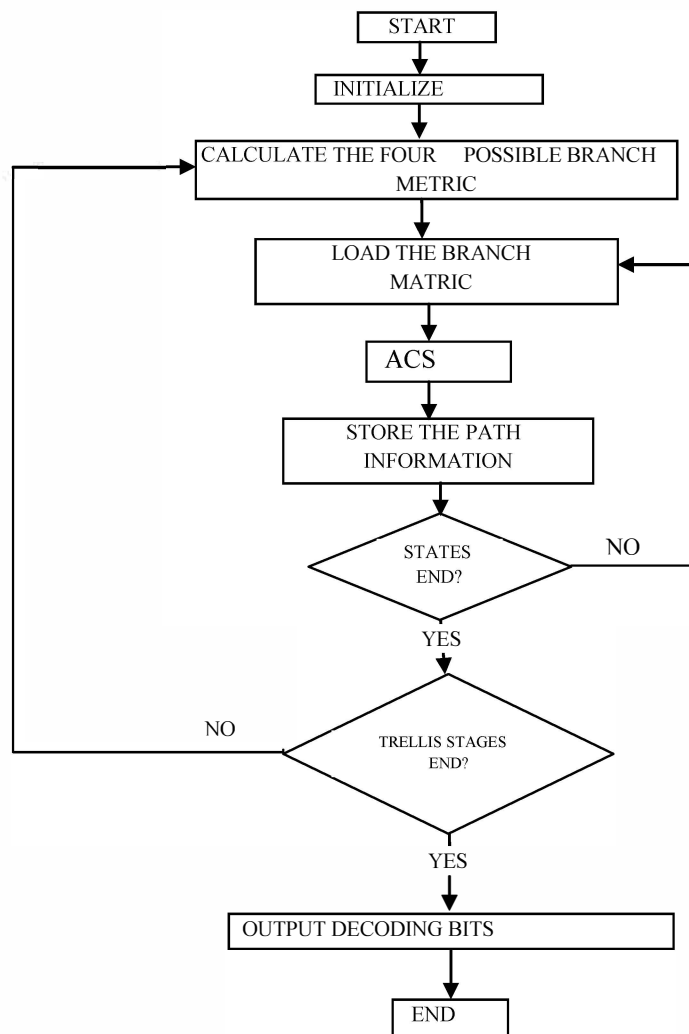


Fig. 3. Flowchart of Viterbi Decoder.

V. RESULTS AND IMPLEMENTATION OF CONVOLUTION ENCODER AND VITERBI DECODER

The Convolution Encoder and Viterbi Decoder have been implemented in Altera DE board. The device taken for the implementation is EP4SGX70HF35C2 of Stratix IV family and design has been simulated using ModelSim 6.6d. Table I shows the hardware utilization summary of Convolution Encoder while table II shows the hardware utilization of Viterbi Decoder in terms of ALUTS and logic registers. From table I, it is evident that there is improvement of proposed implementation in terms

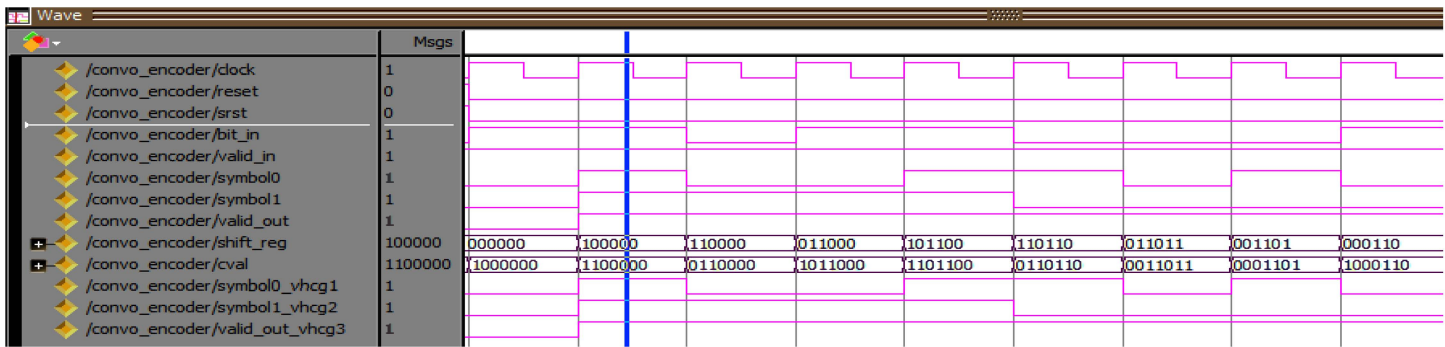


Fig. 4. Simulation Result of Convolution Encoder obtained from EP4SGX70HF35C2 device

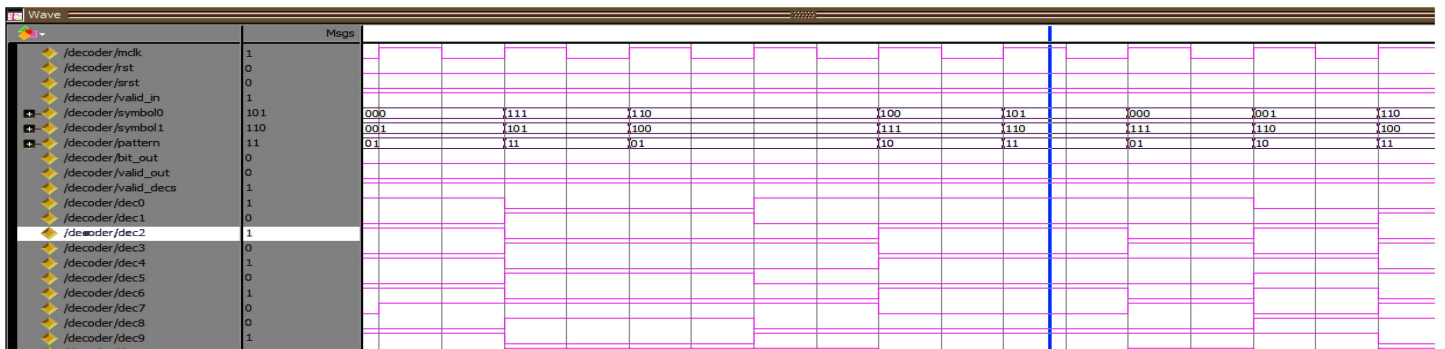


Fig. 5. Simulation Result of Viterbi Decoder obtained from EP4SGX70HF35C2 device

of area, compared to the implementation done in [6]. Figure 4 shows the simulated outputs of Convolution Encoder and figure 5 shows the outputs of Viterbi Decoder which are obtained from FPGA using EP4SGX70HF35C2 device of Stratix IV family.

TABLE I. RESULTS OF CONVOLUTION ENCODER IN TERMS OF HARDWARE UTILIZED

Device EP4SGX70HF35C2	Used	Available	Utilization
# of ALUTS	4	58080	< 1%
# of dedicated logic register	9	58080	< 1%
Total pins	8	584	1%

TABLE II. RESULTS OF VITERBI DECODER IN TERMS OF HARDWARE UTILIZED

Device EP4SGX70HF35C2	Used	Available	Utilization
# of ALUTS	7991	58080	14%
# of dedicated logic register	2523	58080	4%
Total pins	14	584	2%
# of block memory bits	131072	6617088	2%

VI. CONCLUSIONS

We have implemented area optimized VLSI architecture of Convolution Encoder and Viterbi Decoder. The proposed architecture is implemented in EP4SGX70HF35C2 device of Stratix IV family and realized using ModelSim 6.6d. It shows area improvement in terms of hardware utilized. The silicon validation is done by programming the FPGA with VHDL code of the proposed structure implementation.

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